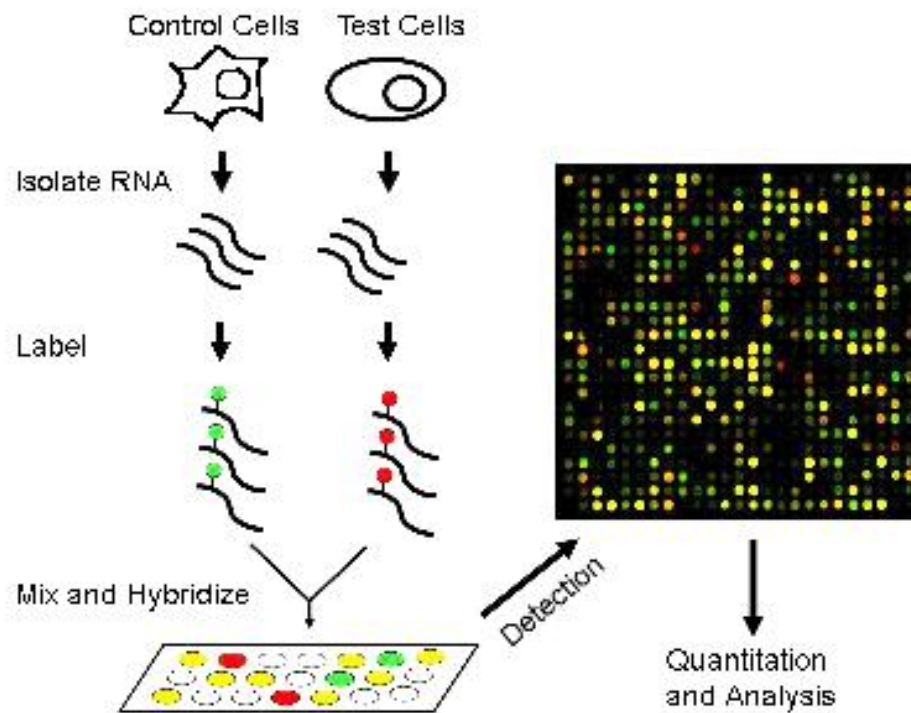


Computational Biology

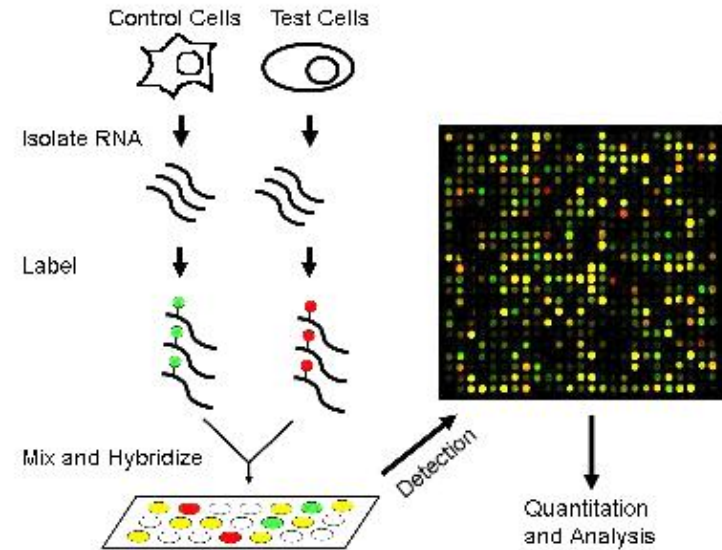
Lecture 2: *Protein Expression*



Microarrays, proteomics, protein interaction

Outline

1. **Protein Expression**
2. Correlations and Causality



Protein Interactions

- Cells are complex organisms where proteins work together to solve problems
- The concentration of proteins will vary depending on what the cell is doing
- The concentration of one protein will affect (excite or inhibit) the expression of other proteins
- This lecture is about try to find these interactions
- We will later look at the interactions in detail

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- The direct measure of protein concentration is known as **proteomics**
- It is a big subject in itself
- Chemically there is no easy way to read off a protein sequence
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Measuring mRNA Levels

- An indirect method to measure protein concentration is to measure the concentration of mRNA
- Although not a perfect proxy it does provide a reasonable approximation to the protein concentration
- It is much easier and cheaper than direct measurement of protein concentration
- Two methods
 - ★ Microarrays (used for the last 20 years)
 - ★ RNA sequencing (beginning to take over)

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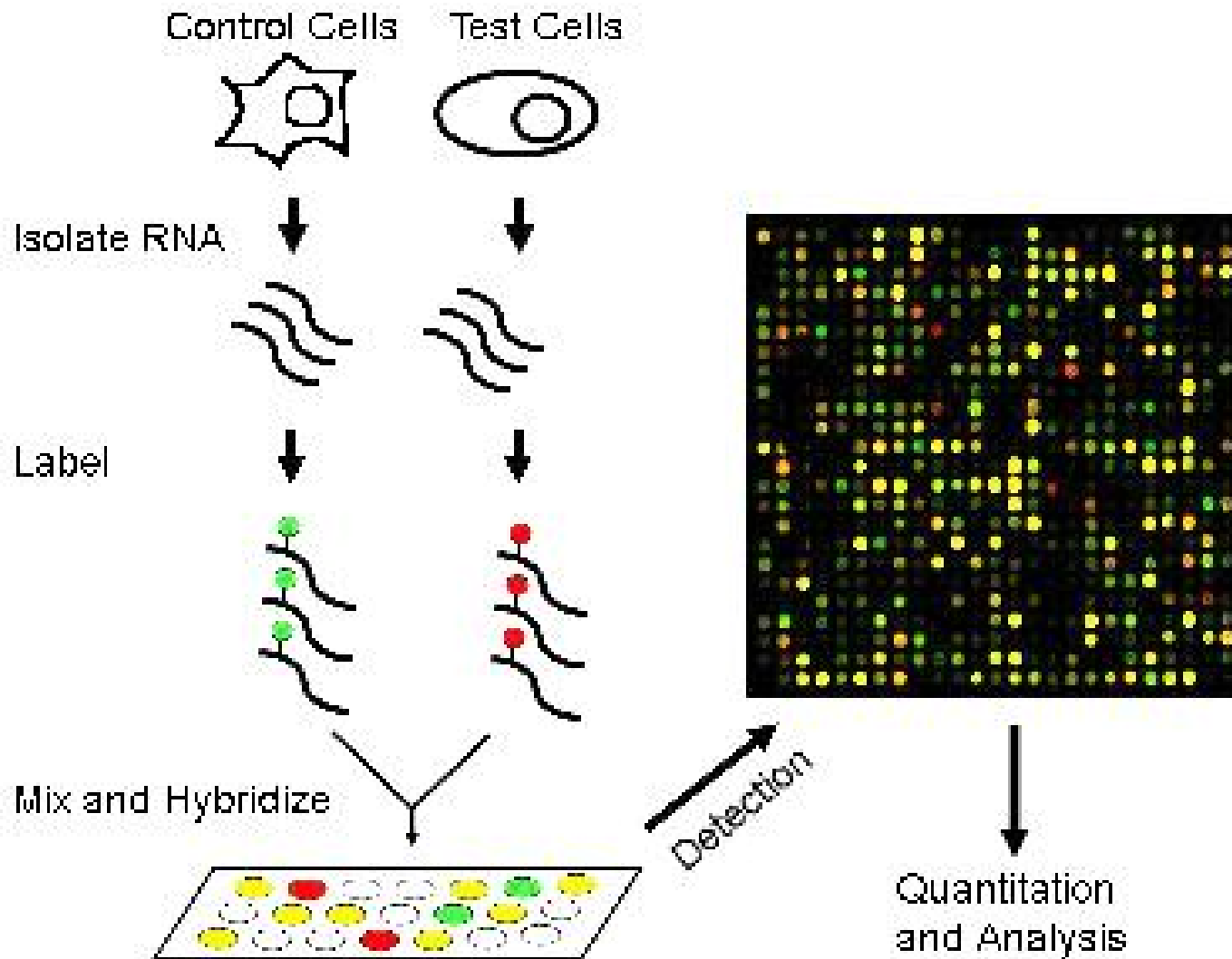
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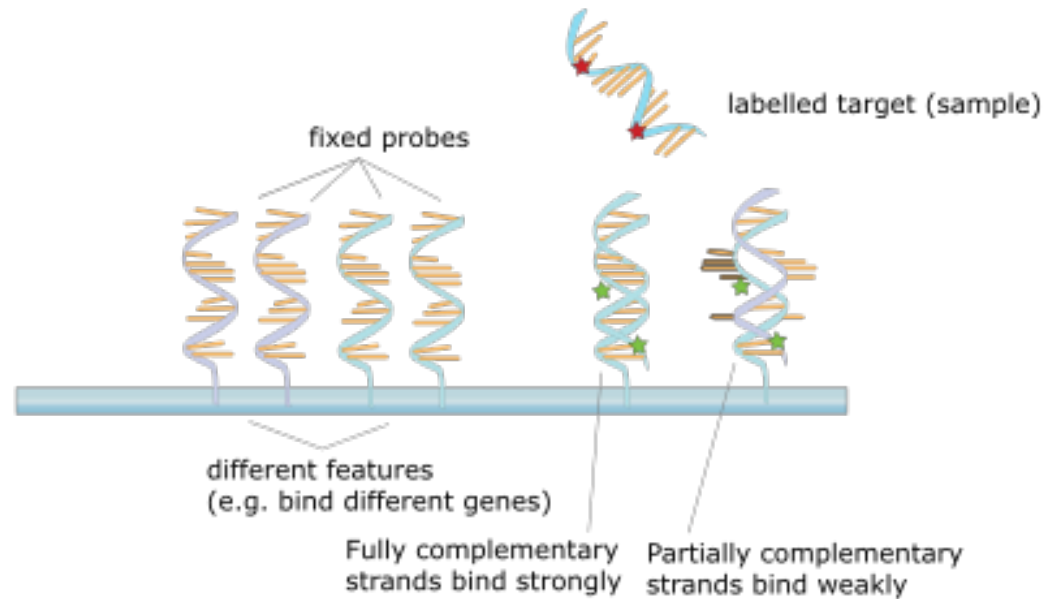
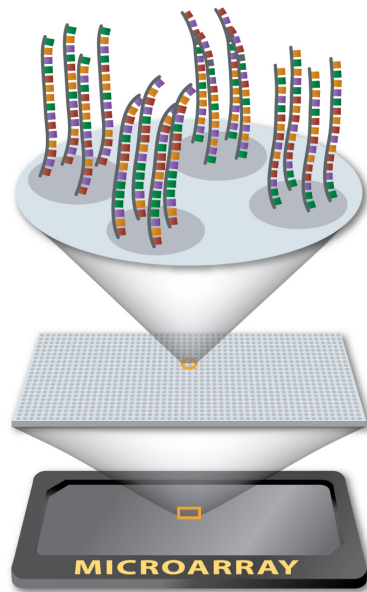
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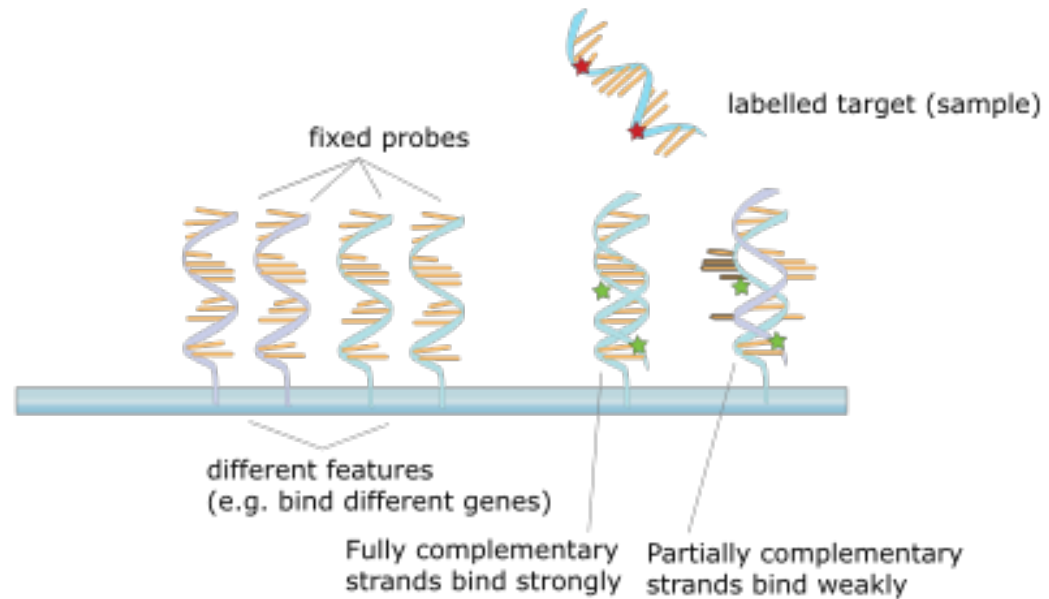
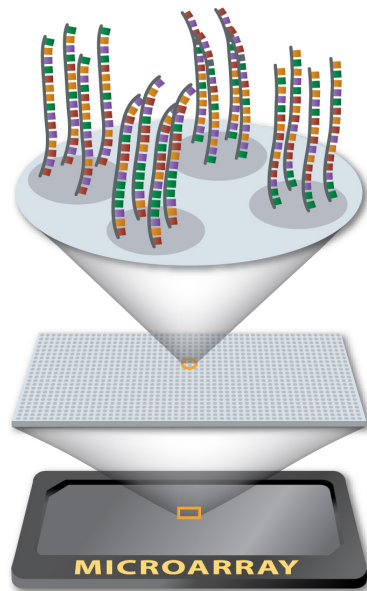
DNA Microarrays

- RNA turned into cDNA
- Micro-wells contain complementary strands of DNA



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DNA Microarrays

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- The cDNA will bind to a complementary strand attached to a microarray
- The microarray needs to be built with the correct DNA strands for the organism being used
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- Results show over-expression (red) or under-expression (green) relative to a group of control cells
- Study cell that is doing something different to the control
- Always use a population of cells so we are only looking at population averages

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Time Series

- To get an understanding of the correlation between proteins we usually study the evolution over time of the cells
- E.g. add sugar, starve the cell, increase the temperature
- Note, that we are looking at a different population of cells since we kill the cells to extract RNA
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Single Cell RNA-SEQ

- With third generation sequencing technology it is even possible to measure the RNA concentrations in a single cell
- This is expensive, but allows researchers to identify different cells in a population
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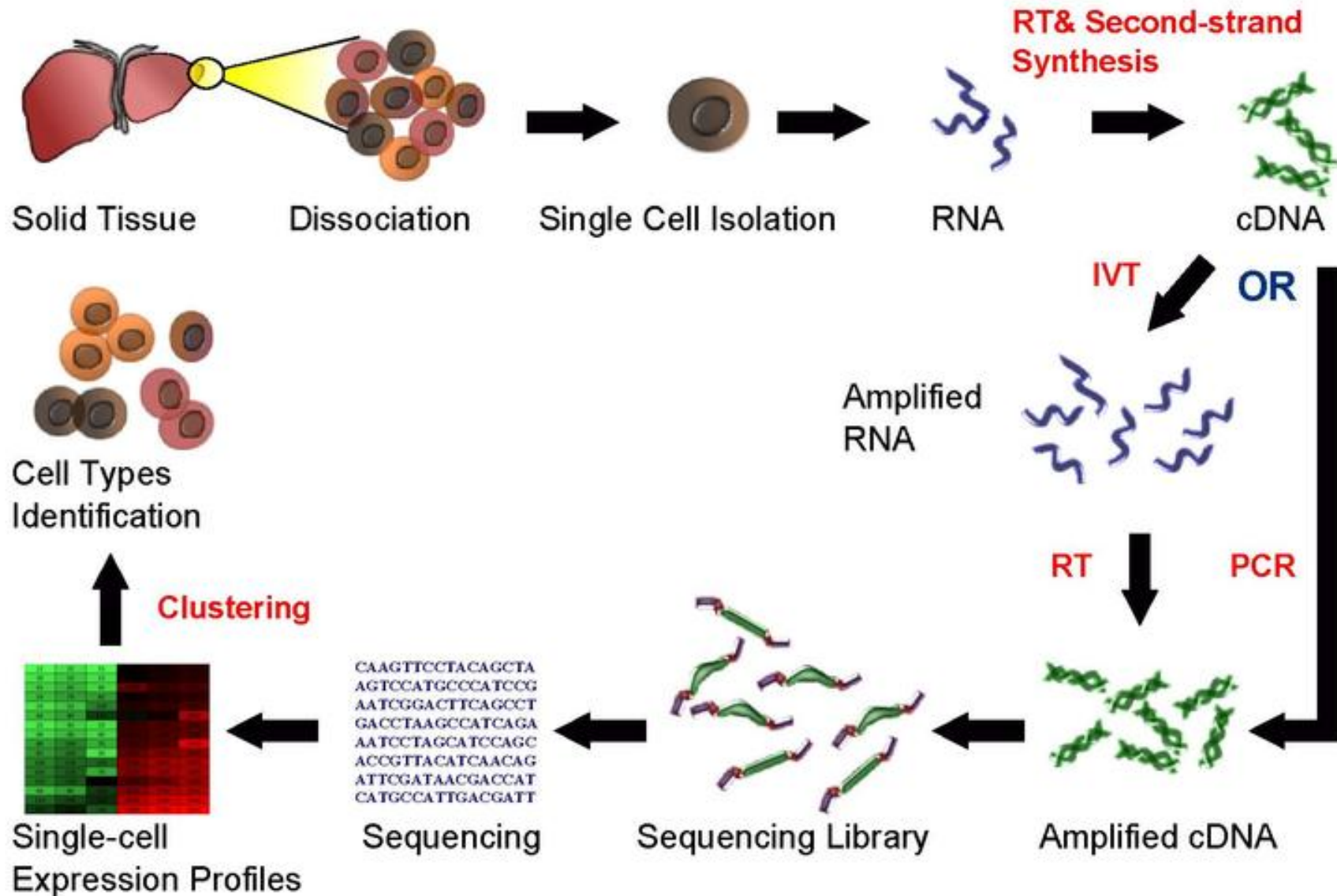
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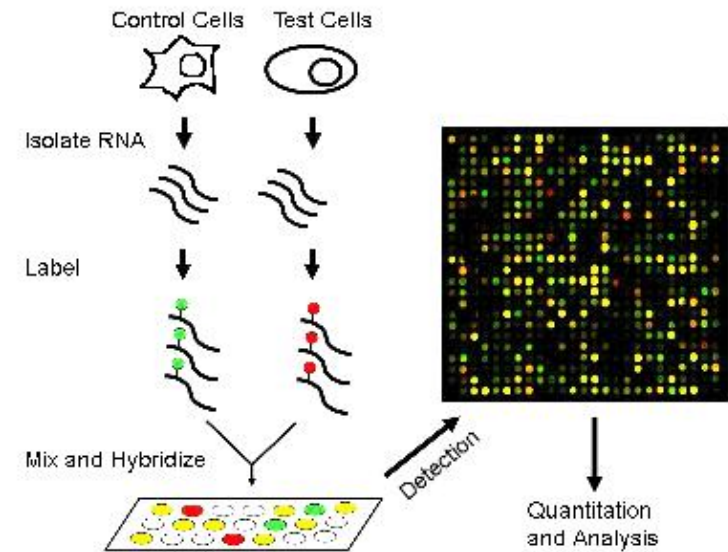
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Single Cell RNA Sequencing Workflow



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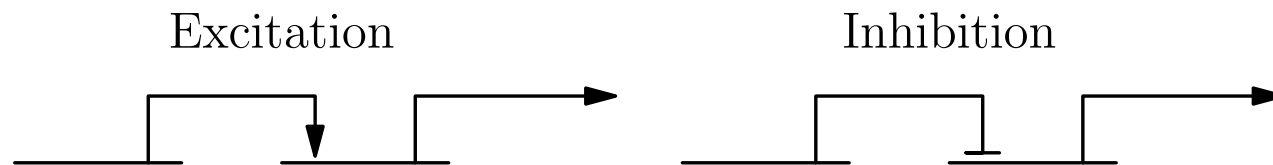


Correlation Between Time Series

- We suppose we have carried out an experiment where we have put a population of cells in a saline solution
- For each mRNA we have a time series (e.g. taken every 30min)

$$\mathbf{X}_i = (X_i(1), X_i(2), \dots, X_i(T))$$

- We can immediately measure if two mRNAs are correlated or anti-correlated

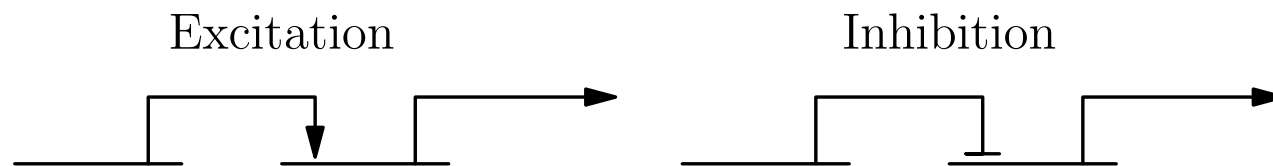


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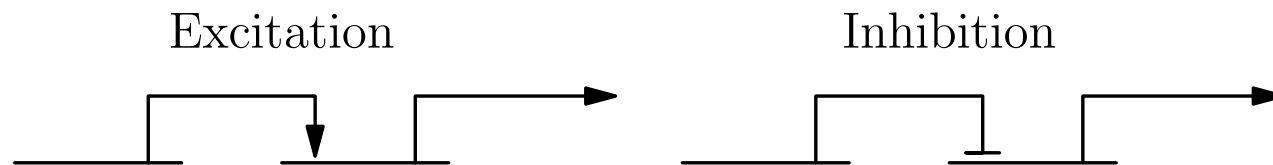


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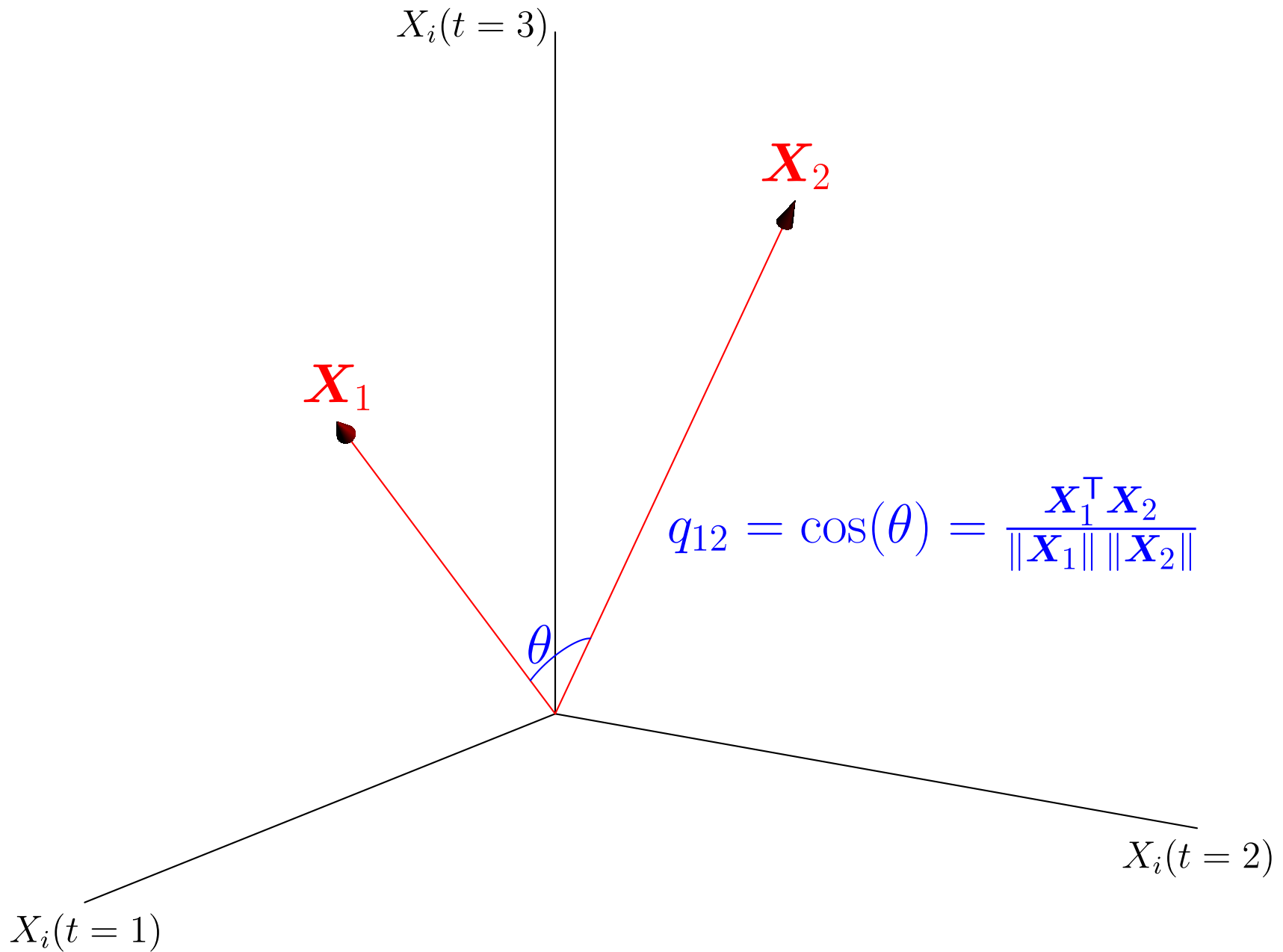
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Pearson Correlation



Correlation and Causality

- The famous mantra is

Correlation does not imply causality

- If two random variables X and Y are correlated then
 - ★ X could affect Y
 - ★ Y could affect X
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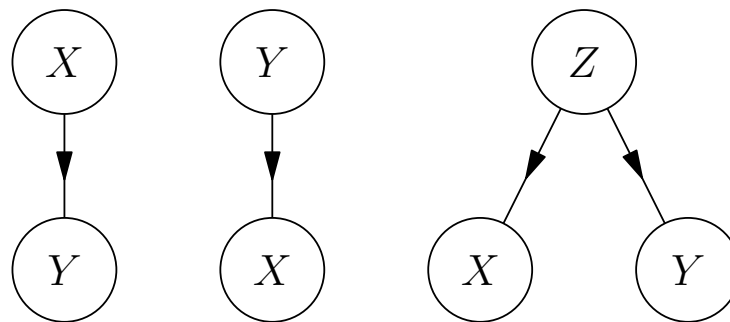
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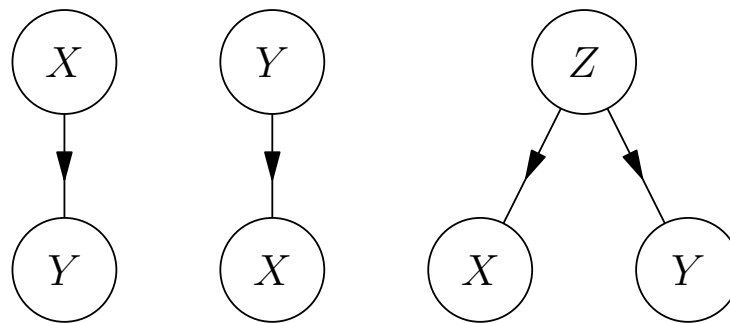
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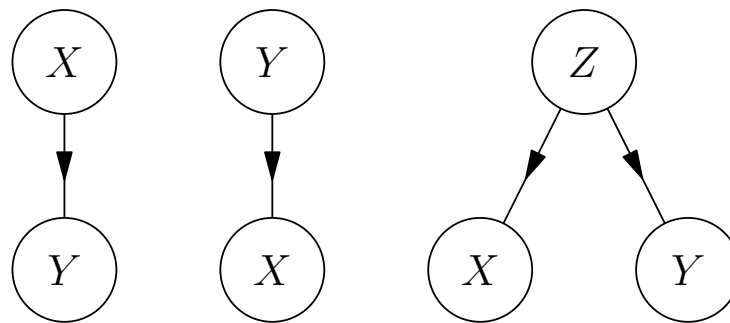
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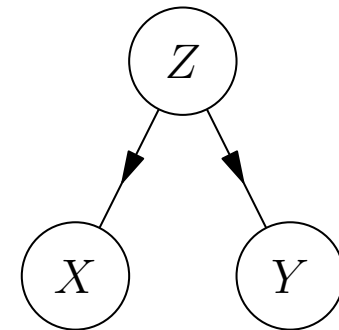
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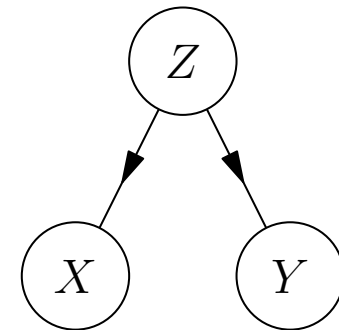


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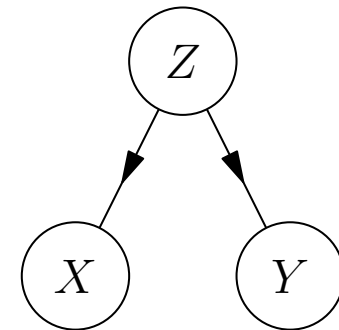


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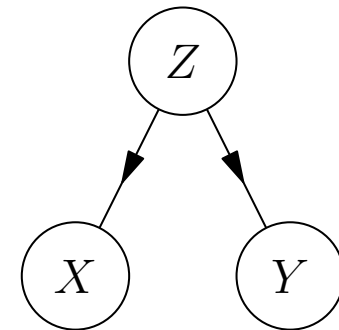


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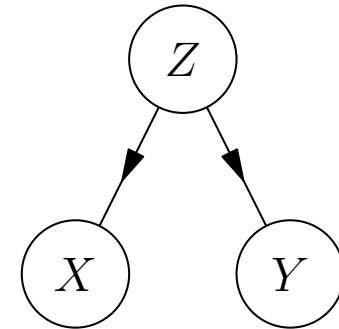


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- If we suppose that the expression data at different times is normally distributed
- One way to find conditionally independent variables would be to measure the covariance between the RNA levels
 $\text{Cov}[X_i, X_j] = \Sigma_{ij}$ and invert the matrix Σ
- But we can only estimate the empirical covariance

$$\hat{\Sigma}_{ij} = \frac{1}{T-1} \sum_{t=1}^T (X_i(t) - \hat{\mu}_i) (X_j(t) - \hat{\mu}_j), \quad \hat{\mu}_i = \frac{1}{T} \sum_{t=1}^T X_i(t)$$

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- Given our data $\mathcal{D} = (X_i(t) | i = 1, 2, \dots, n \quad t = 1, 2, \dots, T)$ we can try to find the value of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ that maximises the likelihood of the data

$$\begin{aligned}\mathbb{P}(\mathcal{D} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) &= \prod_{t=1}^T \mathcal{N}(\mathbf{X}(t) | \boldsymbol{\mu}, \boldsymbol{\Sigma}) \\ &= \left(\frac{1}{(2\pi)^{d/2} |\boldsymbol{\Sigma}|^{1/2}} \right)^T \prod_{t=1}^T e^{-\frac{1}{2} (\mathbf{X}(t) - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{X}(t) - \boldsymbol{\mu})}\end{aligned}$$

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- The maximum of the likelihood will also be the maximum of the log-likelihood $\mathcal{L} = \log(\mathbb{P}(\mathcal{D}|\boldsymbol{\mu}, \boldsymbol{\Sigma}))$

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- If we maximise with respect to $\boldsymbol{\mu}$ (i.e. solving $\nabla_{\boldsymbol{\mu}} \mathcal{L} = 0$) we find $\boldsymbol{\mu}^* = \hat{\boldsymbol{\mu}}$
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Regularisation

- The maximum likelihood value of the covariance is (solving for $\partial\mathcal{L}/\partial\mathbf{\Sigma} = 0$) given by $\mathbf{\Sigma} = \mathbf{S}$
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- L_1 regulariser are used in regression where it is known as Lasso

$$L = \sum_{k=1}^m (\mathbf{w}^\top \mathbf{x}_k - y_k)^2$$

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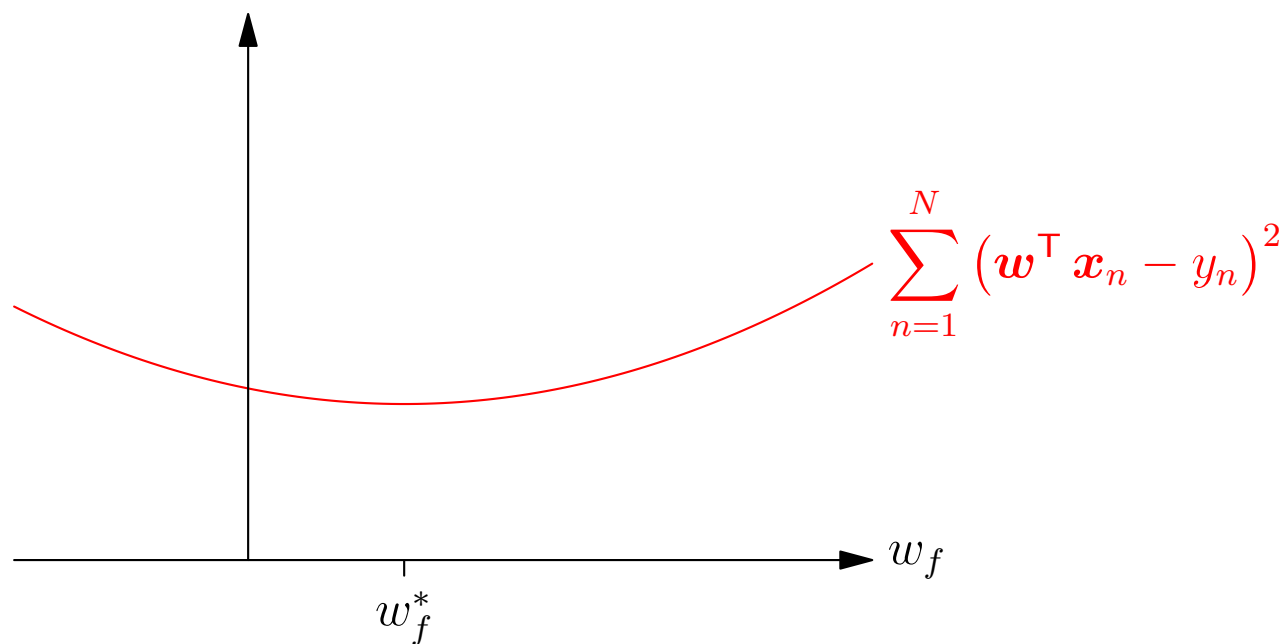
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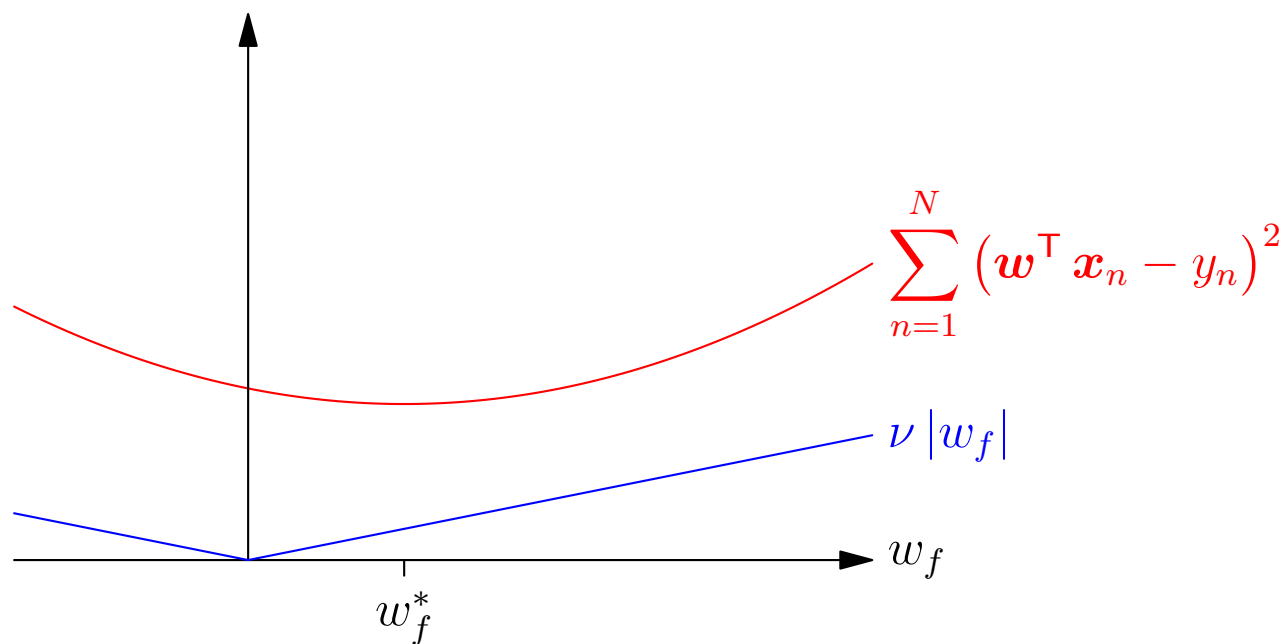


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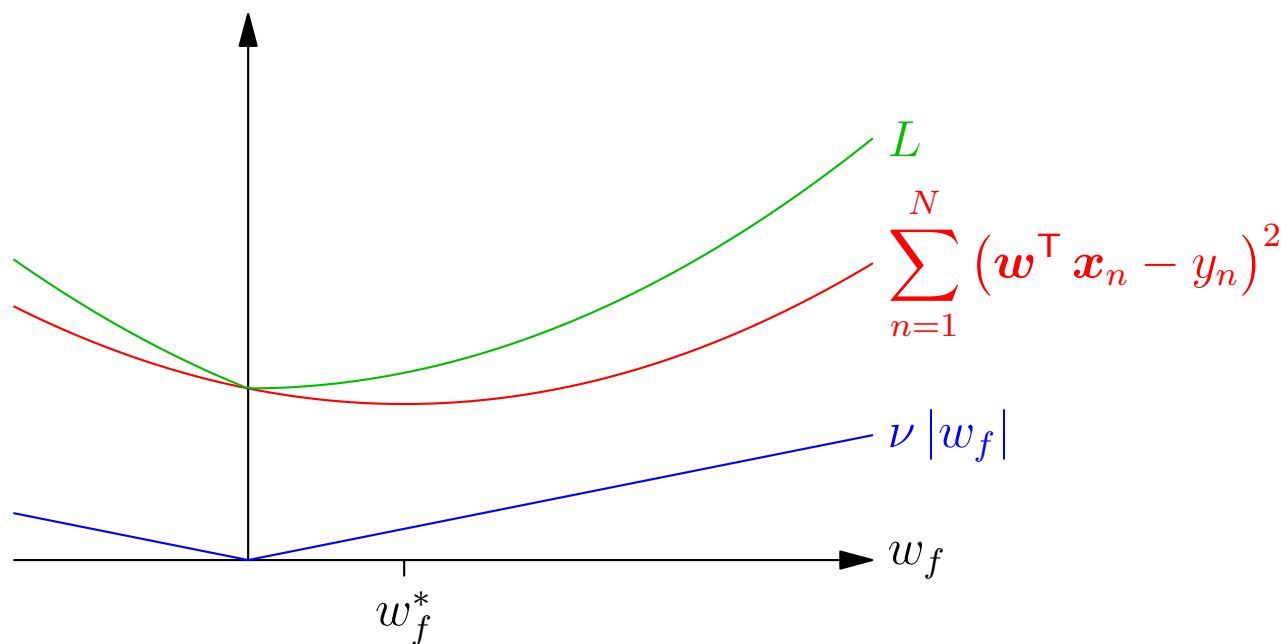


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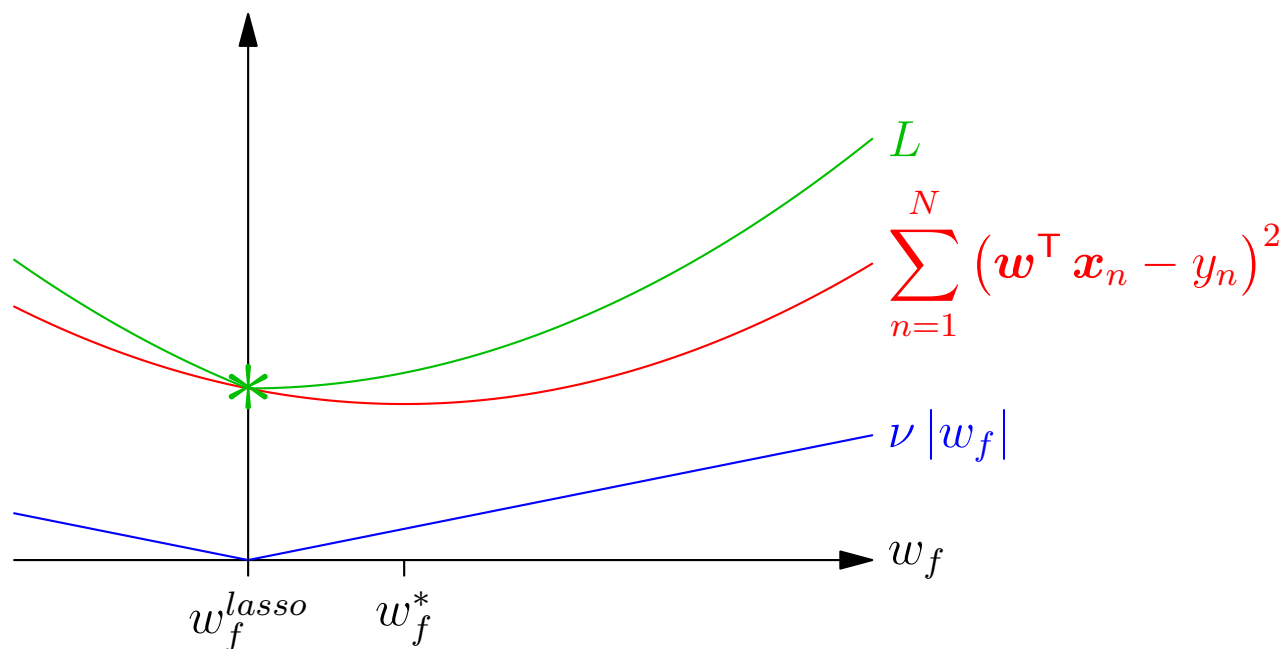


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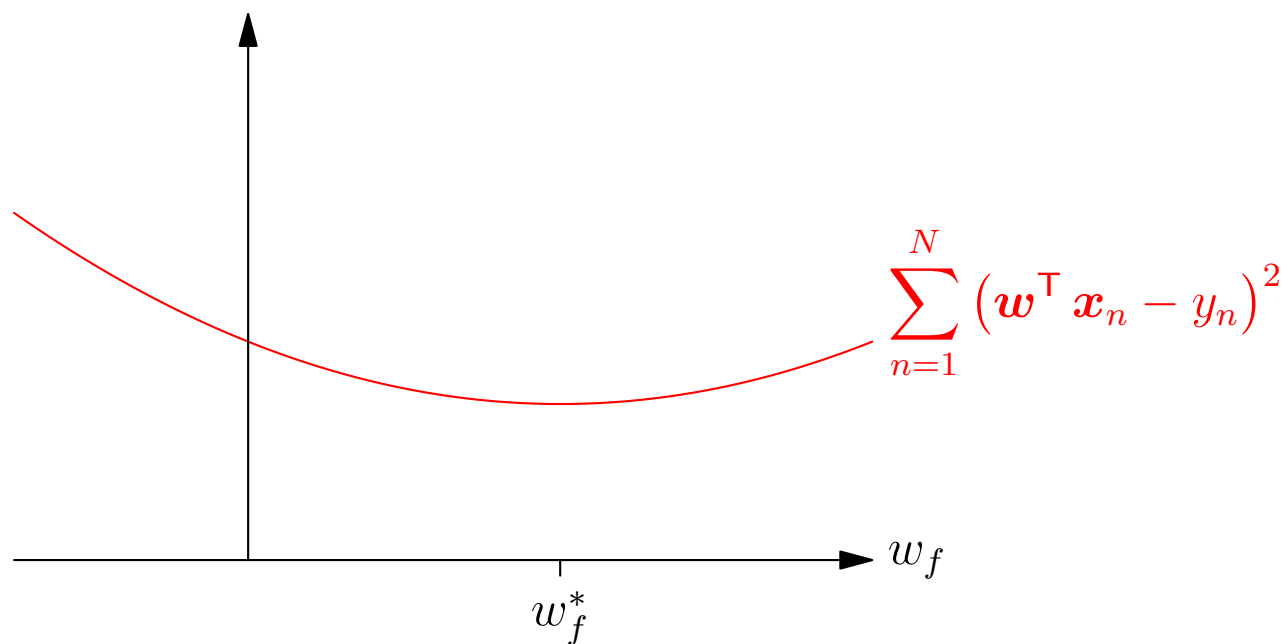


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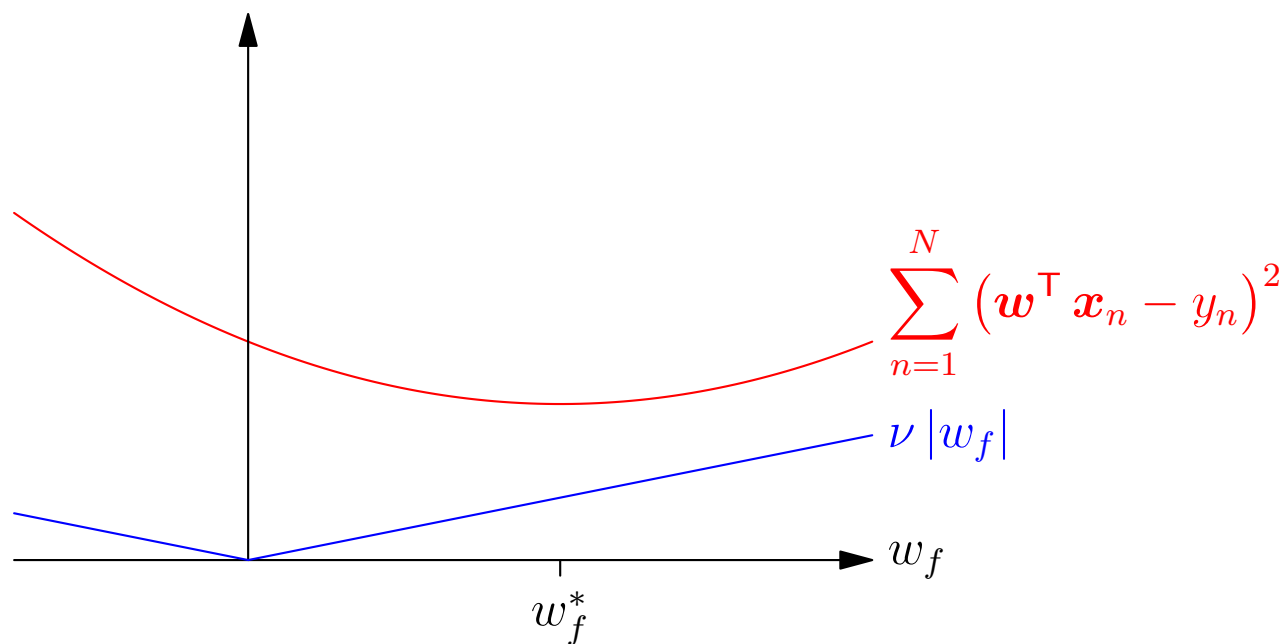


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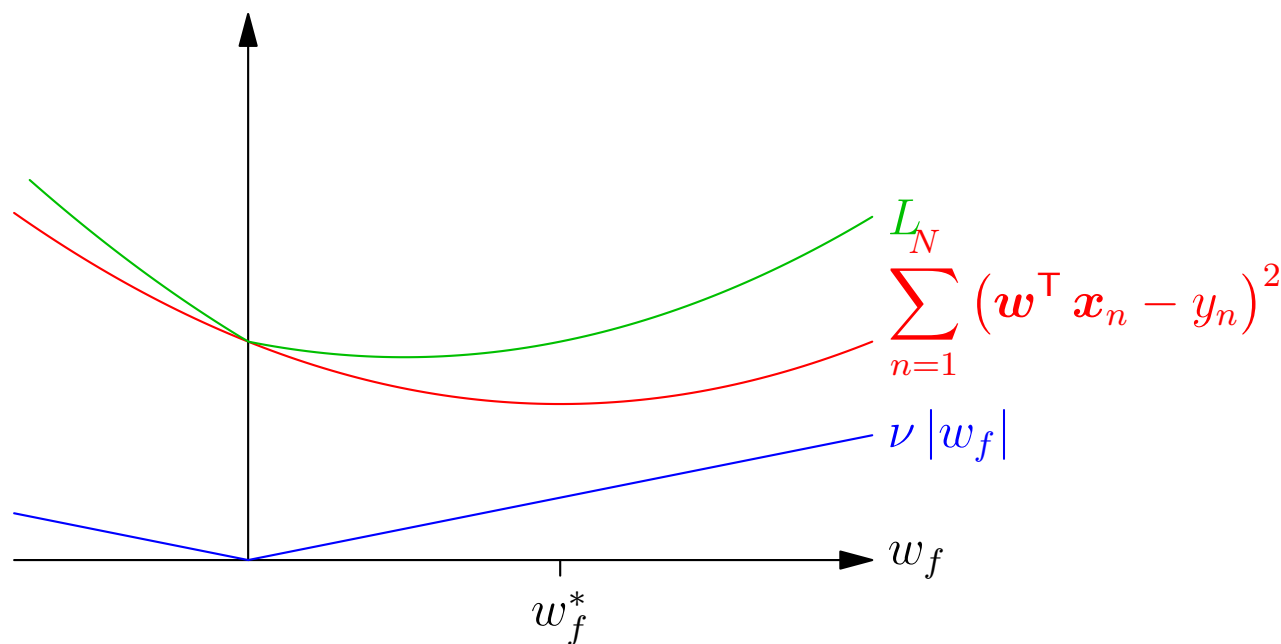


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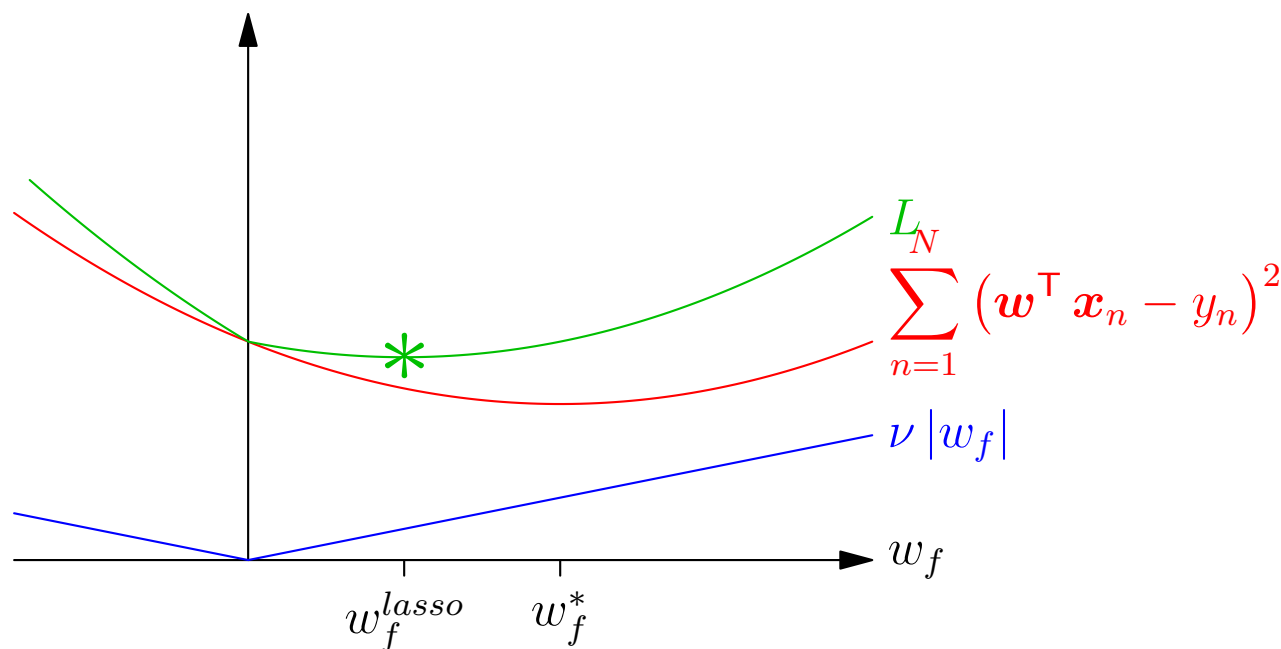


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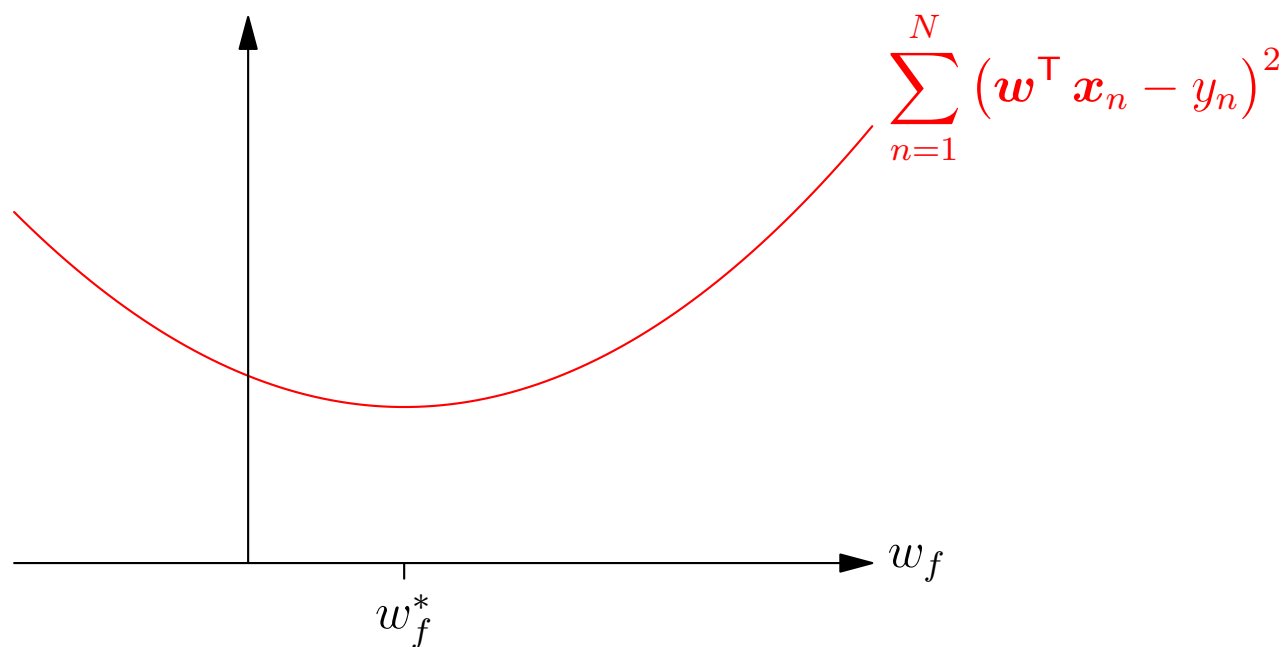


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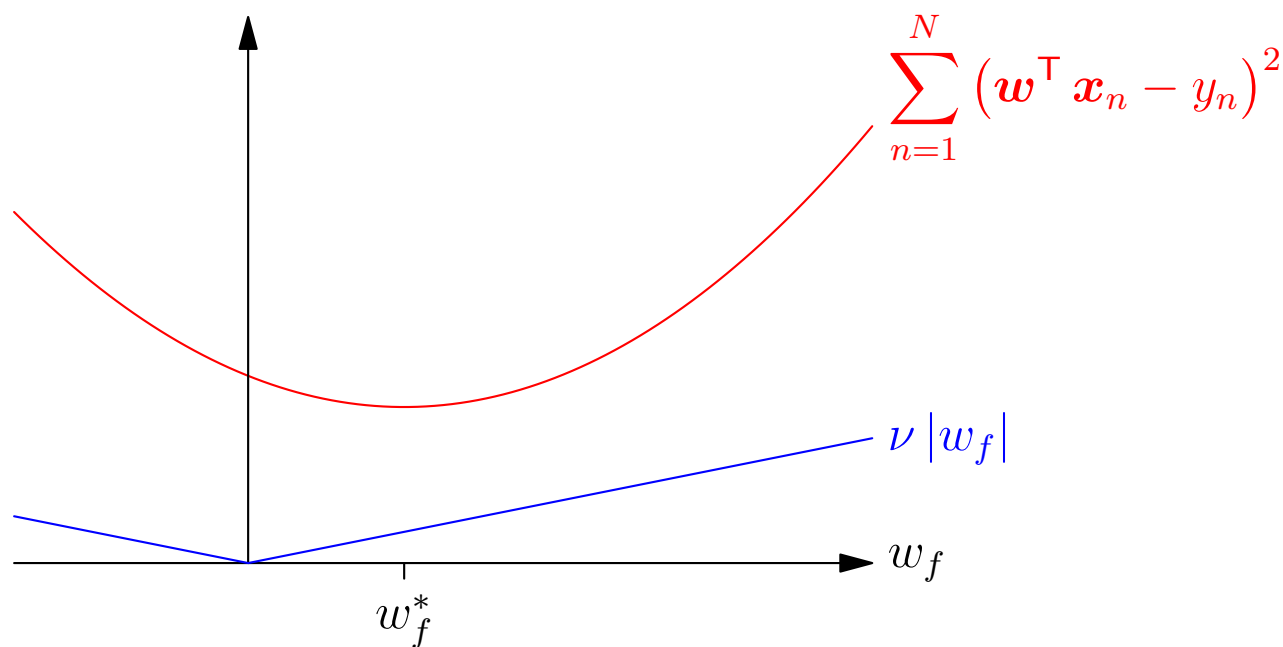


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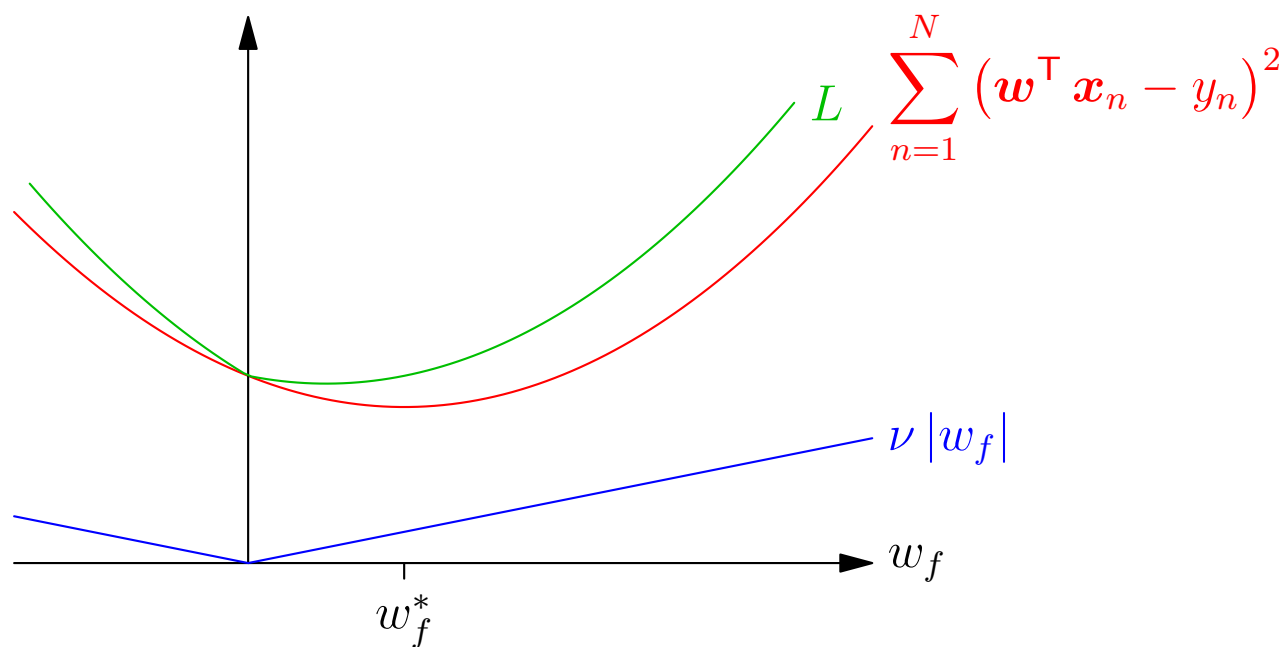


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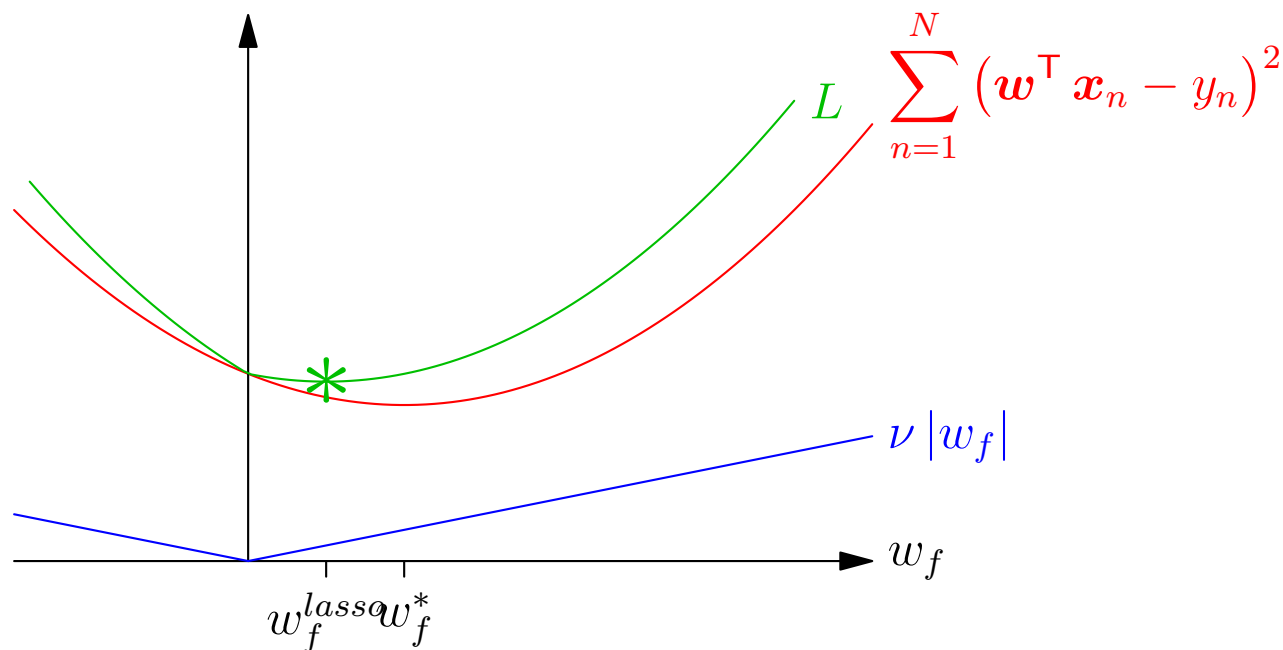


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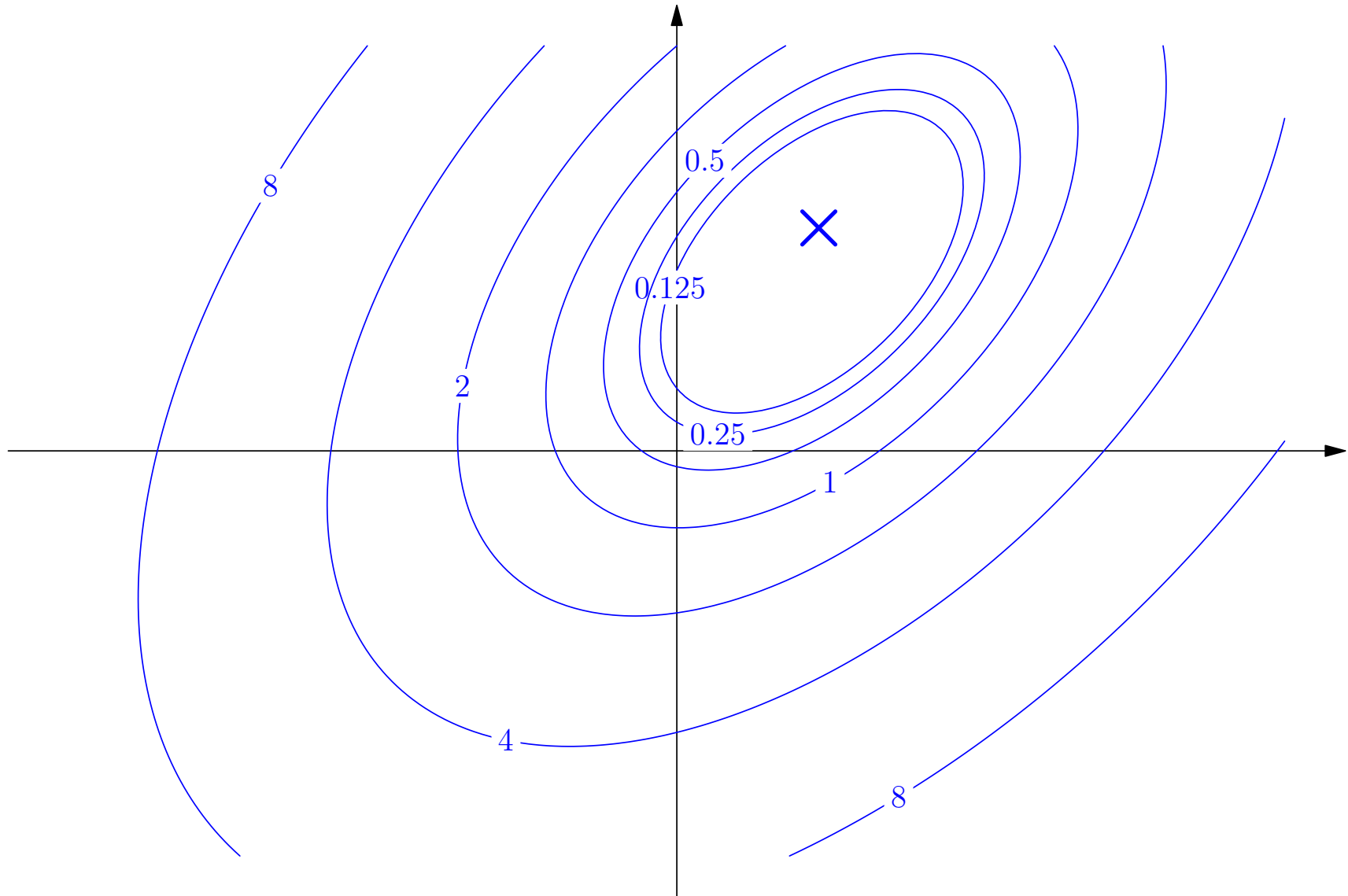
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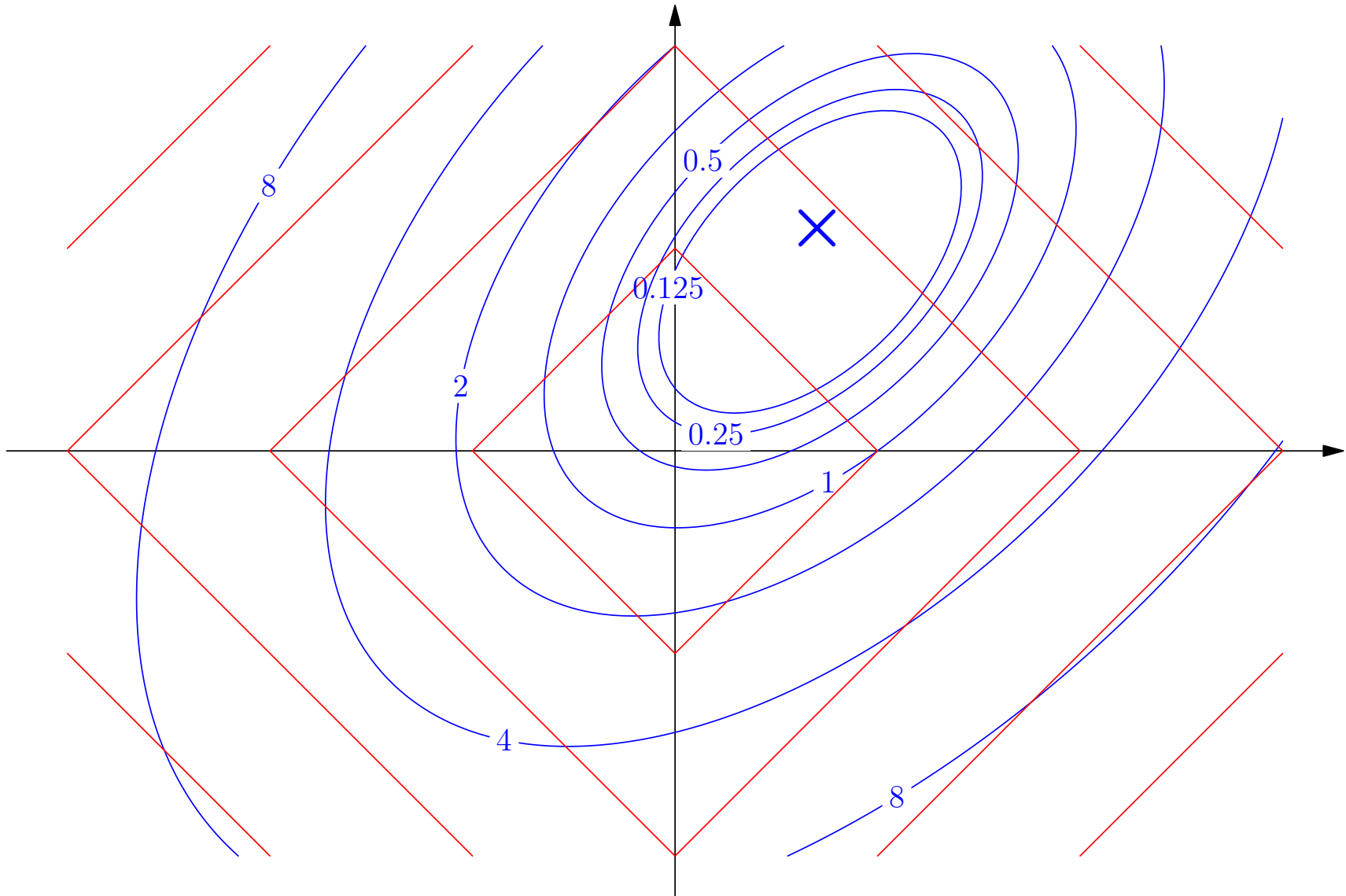
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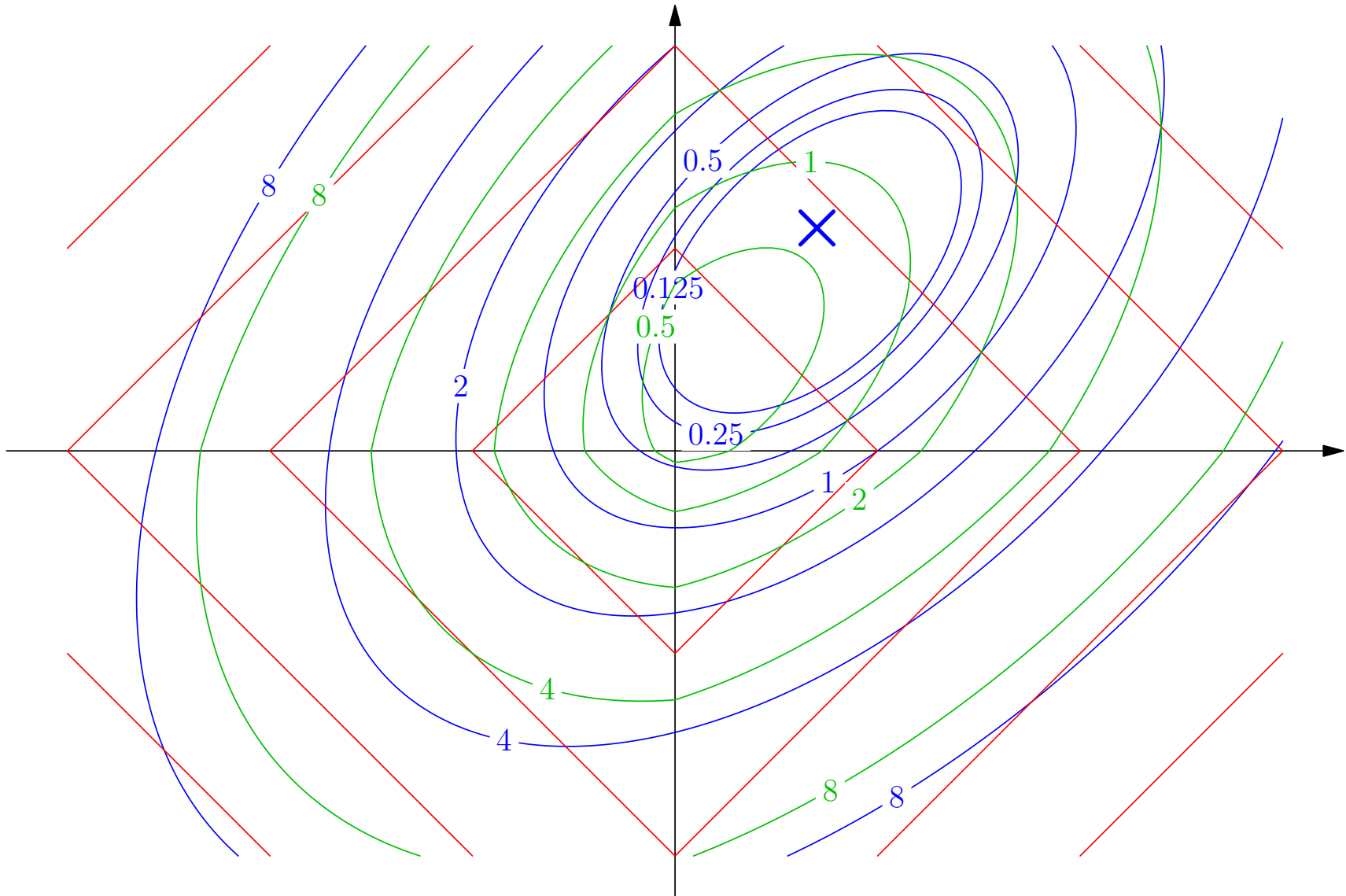
Graphical Lasso



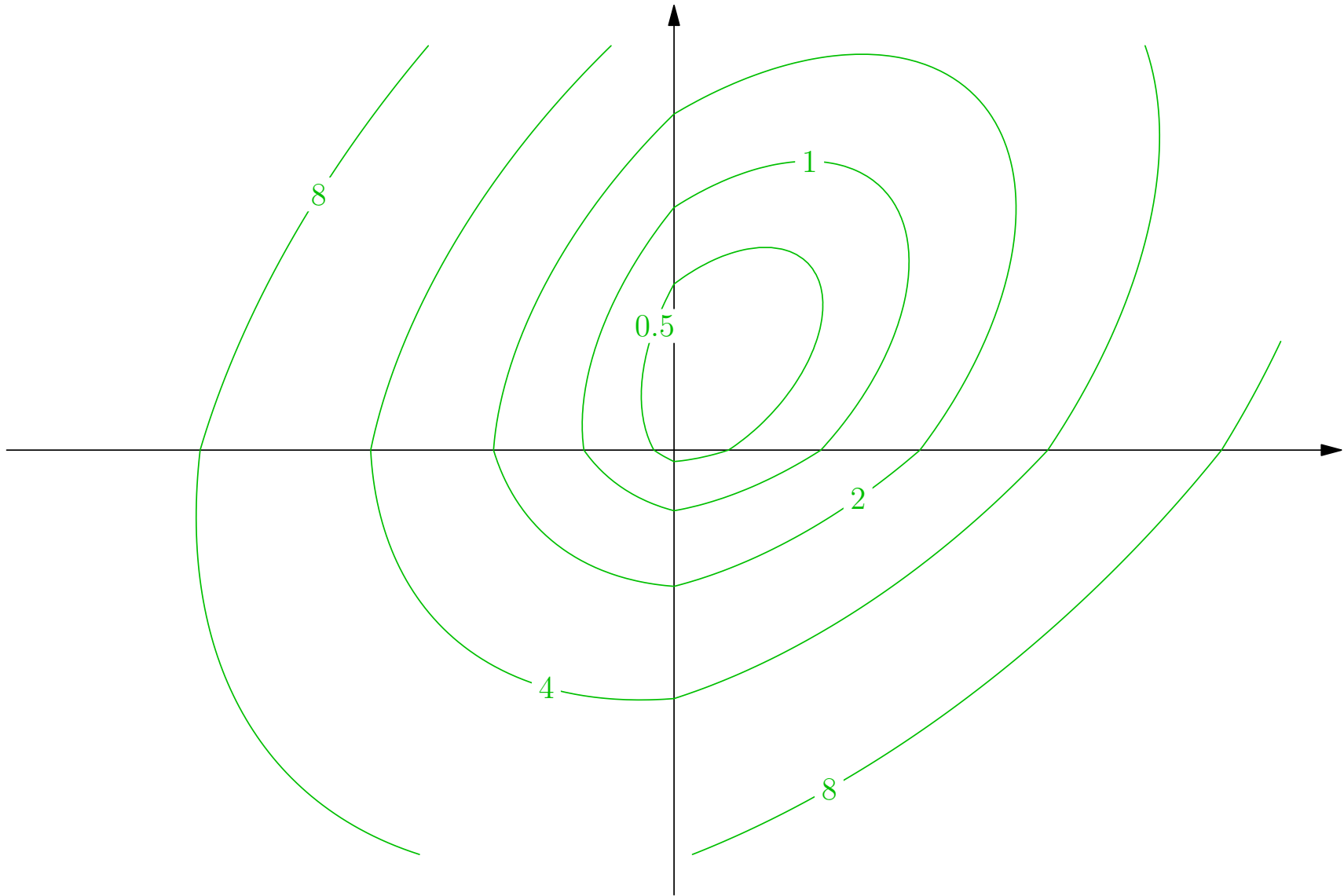
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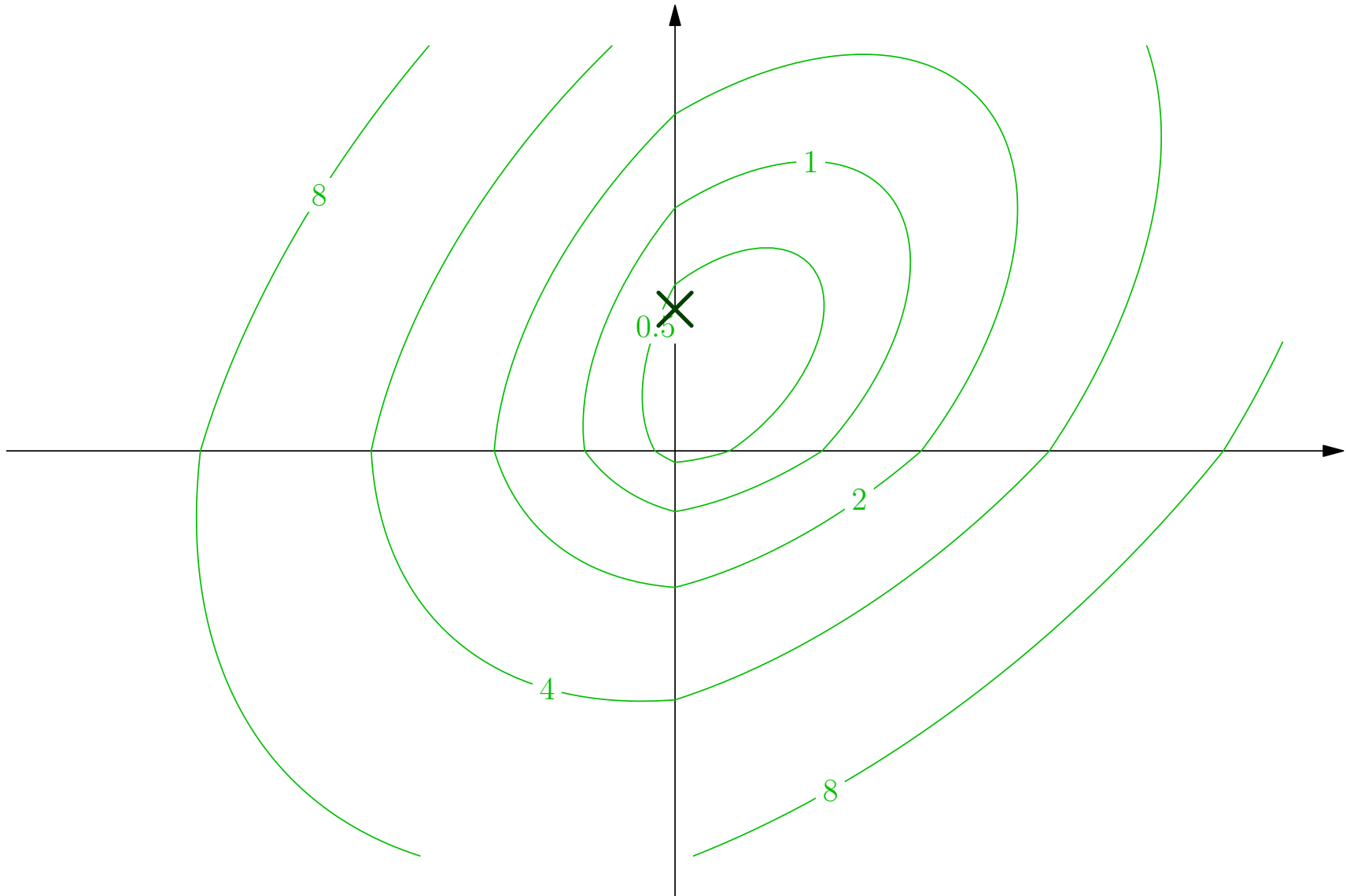
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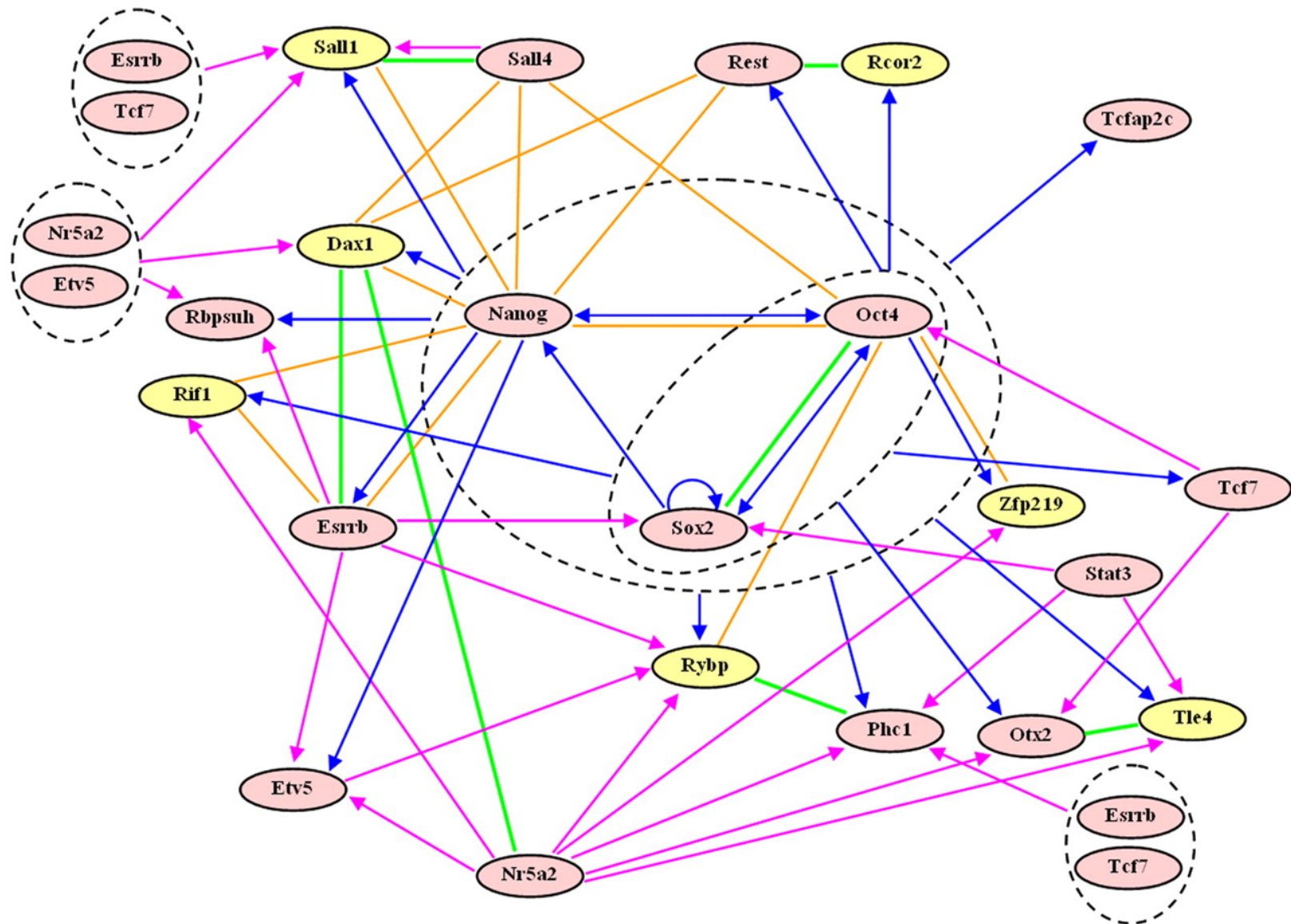
Graphical Lasso



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Gene Regulation Network



Summary

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- By measuring the expression over time we can measure correlations between proteins
- To infer a sparse graphical model that explains the observed correlations we can use the “graphical Lasso” to find a sparse inverse correlation matrix
- This is a first step to understand the control mechanisms of a cell

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